Simulation Arena

# Deal Aging & Multi-Quarter Sales Cycles: The Pipeline Hopper

**Sales says quarter 2; the odds from your history are 0.0% for closing then - 0.0% of the value slips past that date, and 100.0% never closes at all. Plan on $0 of this $3.00M deal, not the full amount. That is after allowing for this deal having sat 11 quarters - a deal that had just arrived would be at 0.0%.**

Seed 42 | Run run_9c0f80ec0b56 | 55,069 scenarios tested | Markov chain on your own stage-to-stage… | 21 Sep 2026, 21:33

Same seed, same numbers - this run can be reproduced exactly.

## How this works

Every deal on your board hops from stage to stage. We counted how your own deals actually hopped - forward, staying put for a quarter, or dying - and turned those counts into odds. We also account for how long a deal has been sitting still: the older it gets, the more likely it is to slip or die, so a deal that has been parked for a year is priced differently from one that has just arrived. Then we rolled the odds forward one quarter at a time, which gives the chance of the deal being anywhere on the board in each future quarter. Money is then weighted by those odds: if there is a 1-in-8 chance of closing next quarter, only an eighth of the value deserves to be in next quarter's plan.

## The numbers that matter

CHANCE IT CLOSES IN Q2, AS SALES PROMISED

**0.0%**

A slim chance the money actually lands in the quarter it was promised for.

vs if we ignored how long it has been sitting: 0.0%

CHANCE IT SLIPS PAST THAT QUARTER

**0.0%**

The chance the deal does close, but later than the date sales promised.

CHANCE IT DIES INSTEAD

**100.0%**

The deal never closes - it stalls or the customer walks.

vs 0.0% still stuck in Legal at the end of the horizon

WHAT YOU CAN BANK ON BY Q2

**$0**

The deal value weighted by its real chance of arriving - 0.0% of $3.00M.

CASH COMMITTED AGAINST ODDS

**$3.00M**

You are planning to spend $3.00M that has only a 0.0% chance of arriving.

vs $3.00M promised vs $0 realistic

QUARTERS UNTIL IT REALLY CLOSES

**n/a**

On average, counted from today, for the deals that do close (3.1 quarters to a final yes or no).

vs sales promised 2

HOW LONG THIS DEAL HAS BEEN SITTING

**11 quarters**

The older a deal gets, the more likely it is to slip or die - we have priced that in.

vs we treat it as stale after 4 quarters

## How we got there

### 1. What we measured

We read 55,069 real stage moves from your history and turned them into odds: from each stage a deal moves forward, sits still for a quarter, or dies.

Evidence: 55,069 observed moves

### 2. The forecast versus the odds

Sales has this deal closing in Q2. The odds from your history put it at 0.0% for landing in that quarter - and a 100.0% chance of never landing at all.

Evidence: 0.0% vs 100%

**3. Where the time actually goes**

Deals do not walk through the stages, they sit in them. Looking across the whole horizon, 0.0% of the deals that close do so later than Q2, and 0.0% close after this fiscal year has ended. Even at the end of Q6, 0.0% are still sitting in Legal.

Evidence: 0.0% stuck in Legal

**4. Why the age of the deal matters**

This deal has been running 11 quarters, and we priced it that way. A deal that had only just arrived would show 0.0% for Q2 instead of 0.0%. Deals that sit still are the ones that quietly die, which is why age is part of the forecast rather than an afterthought.

Evidence: 0.0% at 11 quarters vs 0.0% when fresh

**5. The number to plan with**

Weighted by the odds, this deal is worth $0 to Q2's plan, not $3.00M. The difference, $3.00M, is money you would be committing before it has any real chance of arriving.

Evidence: $3.00M of over-commitment

**6. What to do about it**

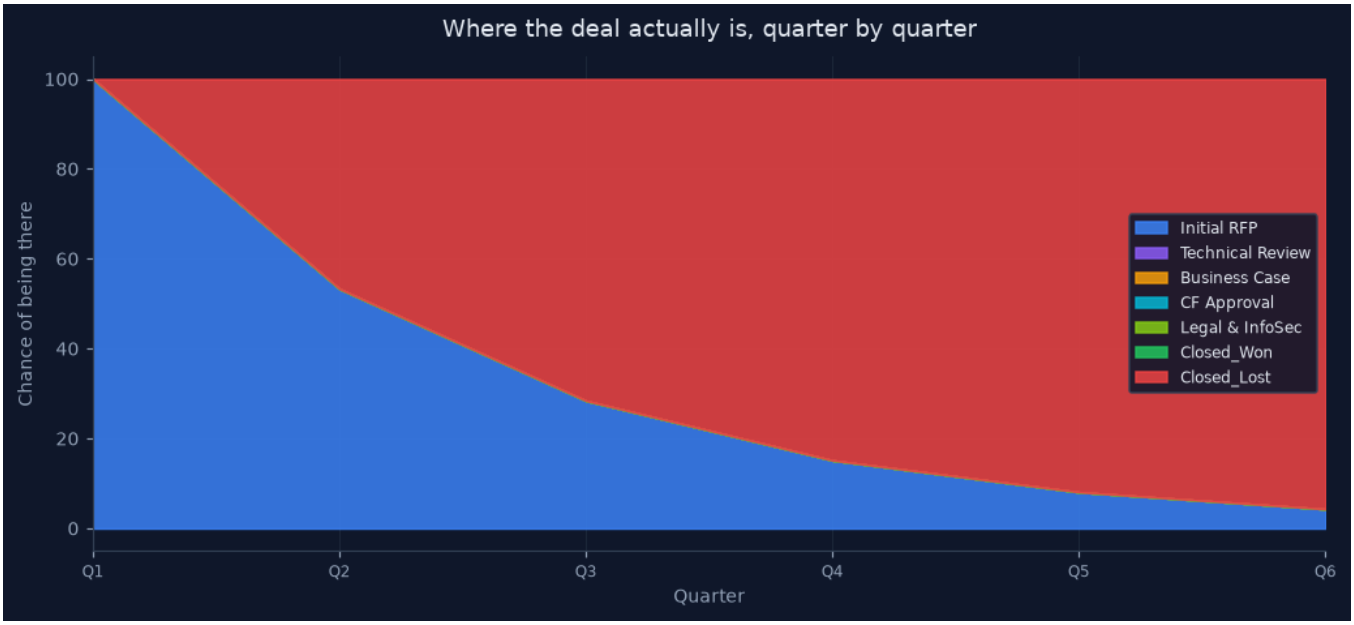
Plan the cash against $0, hold the rest as a contingency, and put the deals that have been sitting the longest in front of whoever owns the stage they are stuck in. A forecast that says 'Q2' with no odds attached is not a forecast - it is a hope.

Evidence: plan on 0.0% of the value

## Charts

### Where the deal actually is, quarter by quarter

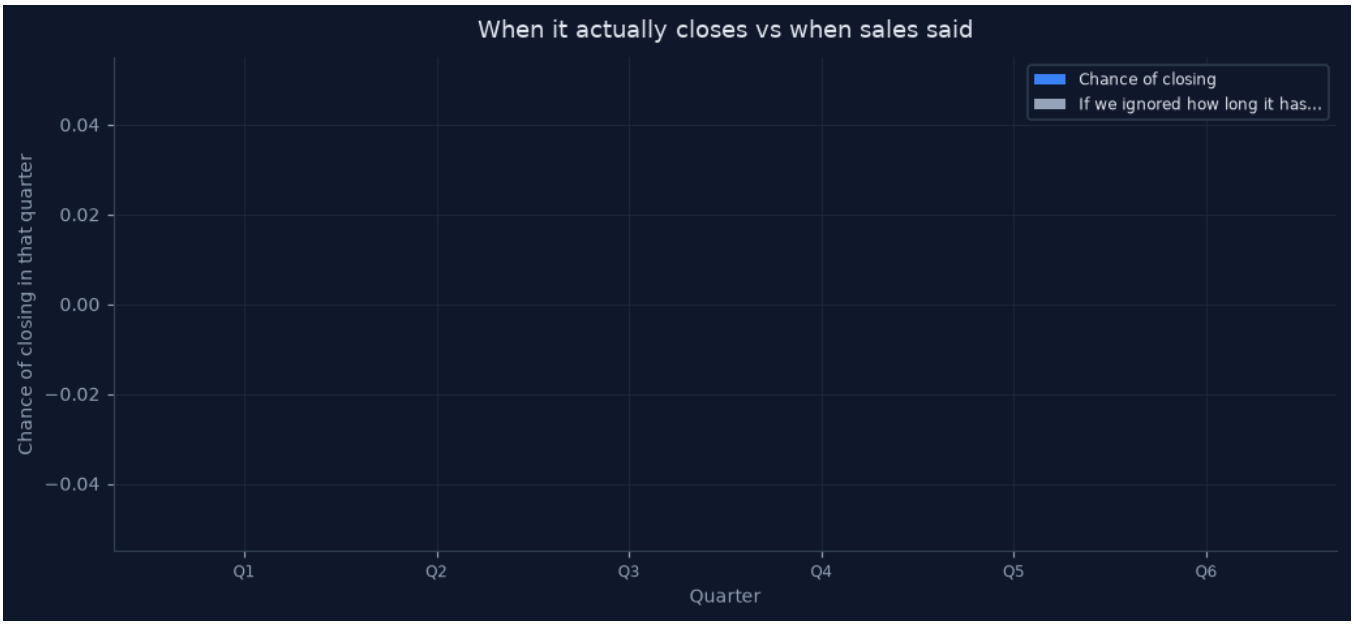
Starting from 'Initial RFP'. Green is won, red is dead.




*What this shows: Read the green band: that is the chance the deal is already closed by each quarter. The grey blocks are the stages it is still sitting in.*

### When it actually closes vs when sales said

Chance of closing in each quarter; sales promised Q2




*What this shows: The tall bar is the quarter the deal is most likely to close in. It is almost never the quarter on the forecast.*

### Money you can actually plan on

Cumulative deal value weighted by its chance of arriving




*What this shows: The gap between the green line and the red line is cash you would be committing before it has arrived.*

## Tables

### Quarter by quarter: where the deal is

The chance of the deal sitting in each stage at the end of every quarter.

| Quarter | Initial RFP | Technical Review | Business Case | CF Approval | Legal & InfoSec | Closed_Won | Closed_Lost |
|---|---|---|---|---|---|---|---|
| Q1 | 53.3% | 0.0% | 0.0% | 0.0% | 0.0% | 0.0% | 46.7% |
| Q2 | 28.4% | 0.0% | 0.0% | 0.0% | 0.0% | 0.0% | 71.6% |
| Q3 | 15.1% | 0.0% | 0.0% | 0.0% | 0.0% | 0.0% | 84.9% |
| Q4 | 8.1% | 0.0% | 0.0% | 0.0% | 0.0% | 0.0% | 91.9% |
| Q5 | 4.3% | 0.0% | 0.0% | 0.0% | 0.0% | 0.0% | 95.7% |
| Q6 | 2.3% | 0.0% | 0.0% | 0.0% | 0.0% | 0.0% | 97.7% |

Read the 'Closed_Won' column to see how the chance of closing builds up.

### The money, weighted by the odds

What this deal is really worth to each quarter's plan.

| Quarter | Chance of closing then | Value earned that quarter | Value banked by then | Chance closed by then | Chance dead by then | Still not banked |
|---|---|---|---|---|---|---|
| Q1 | 0.0% | $0 | $0 | 0.0% | 46.7% | $3.00M |
| Q2 | 0.0% | $0 | $0 | 0.0% | 71.6% | $3.00M |
| Q3 | 0.0% | $0 | $0 | 0.0% | 84.9% | $3.00M |
| Q4 | 0.0% | $0 | $0 | 0.0% | 91.9% | $3.00M |
| Q5 | 0.0% | $0 | $0 | 0.0% | 95.7% | $3.00M |
| Q6 | 0.0% | $0 | $0 | 0.0% | 97.7% | $3.00M |

Row 2 is the quarter sales promised.

### What sitting still costs you

The same deal judged at three ages, plus the age we actually used for it.

| How long it has been sitting | Chance it closes by Q2 | Chance it never closes |
|---|---|---|
| Fresh (under 2 quarters) | 0.03% | 98.4% |
| Settled (2-4 quarters) | 0.02% | 100.0% |
| Stale (over 4 quarters) | 0.0% | 100.0% |
| This deal (11 quarters) | 0.0% | 100.0% |

The older a deal is, the lower it sits. The highlighted row is the age we used for the forecast above.

### Your open deals, weighted the same way (top 25)

Every open deal on the board, priced by the odds of it actually closing - each one judged at its own age.

| Deal | Stage now | Value | Closes next quarter | Within 2 quarters | Ever closes | Expected quarter | Value you can bank on | Quarters sitting |
|---|---|---|---|---|---|---|---|---|
| P-00009825 | Technical Review | $20.00M | <0.01% | <0.01% | <0.01% | n/a | $0 | 10 |
| P-00009825 | Technical Review | $20.00M | <0.01% | <0.01% | <0.01% | n/a | $0 | 10 |
| P-00009825 | Technical Review | $20.00M | <0.01% | <0.01% | <0.01% | n/a | $0 | 10 |
| P-00009825 | Technical Review | $20.00M | <0.01% | <0.01% | <0.01% | n/a | $0 | 10 |
| P-00009825 | Technical Review | $20.00M | <0.01% | <0.01% | <0.01% | n/a | $0 | 10 |
| P-00009825 | Technical Review | $20.00M | <0.01% | <0.01% | <0.01% | n/a | $0 | 10 |
| P-00009825 | Technical Review | $20.00M | <0.01% | <0.01% | <0.01% | n/a | $0 | 10 |
| P-00009825 | Technical Review | $20.00M | <0.01% | <0.01% | <0.01% | n/a | $0 | 10 |
| P-00009825 | Technical Review | $20.00M | <0.01% | <0.01% | <0.01% | n/a | $0 | 10 |
| P-00009825 | Technical Review | $20.00M | <0.01% | <0.01% | <0.01% | n/a | $0 | 10 |
| P-00009825 | Technical Review | $20.00M | <0.01% | <0.01% | <0.01% | n/a | $0 | 10 |
| P-00009673 | Technical Review | $17.09M | 0.02% | 0.03% | 0.06% | 3.9 | $11k | 2 |
| P-00009673 | Technical Review | $17.09M | 0.02% | 0.03% | 0.06% | 3.9 | $11k | 2 |
| P-00009673 | Technical Review | $17.09M | 0.02% | 0.03% | 0.06% | 3.9 | $11k | 2 |

| Deal | Stage now | Value | Closes next quarter | Within 2 quarters | Ever closes | Expected quarter | Value you can bank on | Quarters sitting |
|---|---|---|---|---|---|---|---|---|
| P-00012463 | Business Case | $13.94M | 0.01% | 0.03% | 3.4% | 9.3 | $470k | 1 |
| P-00012463 | Business Case | $13.94M | 0.01% | 0.03% | 3.4% | 9.3 | $470k | 1 |
| P-00006325 | Initial RFP | $12.44M | 0.01% | 0.02% | 0.05% | 4.3 | $6k | 3 |
| P-00006325 | Initial RFP | $12.44M | 0.01% | 0.02% | 0.05% | 4.3 | $6k | 3 |
| P-00006325 | Initial RFP | $12.44M | 0.01% | 0.02% | 0.05% | 4.3 | $6k | 3 |
| P-00006325 | Initial RFP | $12.44M | 0.01% | 0.02% | 0.05% | 4.3 | $6k | 3 |
| P-00016244 | CF Approval | $11.68M | <0.01% | <0.01% | <0.01% | n/a | $0 | 6 |
| P-00016244 | CF Approval | $11.68M | <0.01% | <0.01% | <0.01% | n/a | $0 | 6 |
| P-00016244 | CF Approval | $11.68M | <0.01% | <0.01% | <0.01% | n/a | $0 | 6 |
| P-00016244 | CF Approval | $11.68M | <0.01% | <0.01% | <0.01% | n/a | $0 | 6 |
| P-00016244 | CF Approval | $11.68M | <0.01% | <0.01% | <0.01% | n/a | $0 | 6 |

Value you can bank on = deal value x its chance of ever closing.

## What we recommend

**Do not book $3.00M for Q2. Book $0 and hold the rest back.**

**Risk High** 100 out of 100 | Confidence: High

**Why**

• The chance of closing in Q2 is 0.0%, not 100%.

• 0.0% of the value arrives later than promised. and 100.0% never arrives at all.

• Planning on the full $3.00M commits $3.00M of cash against a 0.0% chance.

• The deal has been running 11 quarters; at that age a deal is judged 0.0% likely to close by Q2 if it were fresh, so sitting still is costing real odds.

**What to do**

**1.** Put $0 in the Q2 plan and treat the remaining $3.00M as contingent on the deal actually reaching the next stage.

**2.** Attach a stage gate to any hiring, inventory or spend that depends on this deal.

**3.** Re-forecast the whole open pipeline the same way, not just this deal - the same optimism is almost certainly in the rest of the board.

**4.** Ask sales to forecast with odds attached, not dates.

**The trade-off:** Booking less now means a smaller number in this quarter's plan - but it avoids the mid-year freeze that follows a promise that does not arrive.

## What we assumed and what to watch

**Worth knowing before you act**

• Sales says Q2; the model says 0.0%. Committing the full $3.00M on that date is the single biggest risk in this plan.

• 100.0% of deals at this stage never close at all - they are not 'slipping', they are gone. Treat that part of the pipeline as zero.

• This deal has been sitting 11 quarters - past the 4-quarter mark where we start treating a deal as stale. Its chance of closing is 0.0% rather than 0.0% for a fresh deal.

### Assumptions

• Learned how deals hop between stages from 55,069 real stage moves (the quarter column told us what happened first). Paths nobody has taken yet are given a small benefit of the doubt rather than a flat zero chance.

• A deal that is won or dead stays that way - we force those stages to be final even if the history is messy.

• We priced how long deals sit still: past 4 quarters a deal gets progressively more likely to slip or die, up to a cap of 90.0% extra risk. The deal being forecast is treated as 11 quarters old.

• Deal value $3,000,000, starting in 'Initial RFP', 6 quarters ahead, fiscal year = 4 quarters.

• Each quarter a deal moves at most one stage forward; it may sit still or die. Deal size does not change the odds.

• Stage behaviour is assumed stable: the pattern in your history continues.

• Columns of your file we actually used: deal_id, stage, stage_order, quarter, amount, entered_quarter, days_in_stage.

• Deal / opportunity ID -> your 'deal_id' column

• Pipeline stage -> your 'stage' column

• Stage order (1,2,3...) -> your 'stage_order' column

• Quarter of this sighting -> your 'quarter' column

• Deal value -> your 'amount' column

• Quarter the deal entered -> your 'entered_quarter' column

• How long the deal has been sitting (quarters) -> not in your file; not used

• How long it has been sitting (days) -> your 'days_in_previous_stage' column

## Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

| Time | Stage | What happened |
|---|---|---|
| 0.00s | prepare | You chose 'deal_id' for 'Deal / opportunity ID'. |
| 0.00s | prepare | You chose 'stage' for 'Pipeline stage'. |
| 0.00s | prepare | You chose 'stage_order' for 'Stage order (1,2,3...)'. |
| 0.00s | prepare | You chose 'quarter' for 'Quarter of this sighting'. |
| 0.00s | prepare | You chose 'amount' for 'Deal value'. |
| 0.00s | prepare | You chose 'entered_quarter' for 'Quarter the deal entered'. |
| 0.00s | prepare | We could not find a 'How long the deal has been sitting (quarters)' column. We left it out and will work without it. |
| 0.00s | prepare | You chose 'days_in_previous_stage' for 'How long it has been sitting (days)'. |
| 0.00s | prepare | Deal / opportunity ID -> your 'deal_id' column |
| 0.00s | prepare | Pipeline stage -> your 'stage' column |
| 0.00s | prepare | Stage order (1,2,3...) -> your 'stage_order' column |
| 0.00s | prepare | Quarter of this sighting -> your 'quarter' column |
| 0.00s | prepare | Deal value -> your 'amount' column |
| 0.00s | prepare | Quarter the deal entered -> your 'entered_quarter' column |
| 0.00s | prepare | How long the deal has been sitting (quarters) -> not in your file; not used |
| 0.00s | prepare | How long it has been sitting (days) -> your 'days_in_previous_stage' column |
| 1.74s | prepare | Read 75,069 records into the model. |
| 1.74s | simulate | Running the simulation… |
| 1.74s | read | Reading how your deals have moved through the pipeline... |
| 2.67s | prepare | Laid the stages out in the order you gave us. |

| Time | Stage | What happened |
|---|---|---|
| 2.67s | prepare | Learned the hop pattern from 55,069 real stage moves across 20,000 deals. |
| 2.67s | prepare | Deals that sit still get riskier: this one has been going 11 quarters, so we priced it against the stale-deal pattern. |
| 2.67s | simulate | Rolling the odds forward 6 quarters from 'Initial RFP'... |
| 2.67s | summarize | Sales says quarter 2. The odds from your history are 0.0% for closing in that quarter. |
| 3.22s | summarize | Done. Plan on $0 of $3.00M for Q2, not the full amount. |